

Rate Sensitivity and Employment:
A Shift-Share Difference-in-Differences Analysis of
Federal Funds Rate Changes on County-Level Employment

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Abstract

This paper estimates the heterogeneous employment effects of federal funds rate changes across U.S. counties using a shift-share difference-in-differences design. Exploiting variation in predetermined 2002 construction and manufacturing employment shares as an exposure measure, I find that counties with greater concentrations of rate-sensitive industries experience systematically higher employment growth during monetary policy tightening. The contemporaneous interaction coefficient is 0.020 ($p < 0.01$), with the dominant effect materializing at a nine-month lag ($\beta = 0.025$, $p < 0.01$) and a cumulative four-quarter effect of 0.042 ($p < 0.01$). Results are robust across alternative base-year exposure measures and outcome specifications.

1. Introduction

How does monetary policy affect local labor markets? While the aggregate employment effects of Federal Reserve interest rate changes have been studied extensively, less attention has been paid to the *heterogeneous* effects across local economies. Counties differ sharply in their industrial composition, and industries differ substantially in their sensitivity to interest rate changes. Construction and manufacturing, which depend heavily on credit markets and durable goods demand, are plausibly more responsive to monetary policy shifts than sectors such as healthcare or government services. This paper exploits cross-sectional variation in industry composition to identify differential employment effects of changes in the federal funds rate (FFR) across U.S. counties.

This paper estimates whether counties with higher pre-existing shares of rate-sensitive industries, specifically construction and manufacturing, experience systematically different employment outcomes when the Federal Reserve raises or cuts the federal funds rate. The sample covers 2,887 U.S. counties from 2002 to 2019, spanning approximately 155,000 county-quarter observations and two complete interest rate cycles: the post-dot-com recovery and hiking cycle (2002–2007), the Great Recession and near-zero rate environment (2008–2015), and the subsequent normalization (2015–2019). Excluding these cycles would limit the design's power; including both provides for identifying variation from both rate hikes and cuts.

The identification strategy is a shift-share difference-in-differences (DiD) design. Counties are assigned a fixed exposure measure based on their 2002 construction and manufacturing employment share from the County Business Patterns (CBP), which is then used to determine all subsequent monetary policy decisions. This exposure is then interacted with quarterly changes in the federal funds rate. Two-way fixed effects absorb all time-invariant county characteristics and all aggregate macroeconomic shocks. Hence, the interaction coefficient identifies the differential employment response to monetary policy across counties with differing industrial structures.

The main result is that high-exposure counties experience *greater* employment growth when the FFR rises. The contemporaneous interaction coefficient is 0.020 ($p < 0.01$). A distributed lag specification reveals that the dominant effect materializes with a three-quarter lag ($\beta = 0.025$, $p < 0.01$), with a cumulative four-quarter effect of 0.042 ($p < 0.01$). This positive sign is consistent with cyclical amplification: the Federal Reserve typically raises rates during economic expansions, and counties with more construction and manufacturing employment benefit disproportionately from the boom conditions that accompany tightening cycles. These results are robust across alternative base-year exposure measures (CBP 2003, 2013, 2014) and an alternative outcome specification using employment growth rates.

2. Literature Review

A well-established literature documents the aggregate employment effects of monetary policy. Romer and Romer (1989, 2004) show using narrative identification that contractionary monetary policy shocks reduce industrial production and employment. More recently, Coibion (2012) and Nakamura and Steinsson (2018) use high-frequency identification of Fed policy surprises to estimate effects on real activity, finding significant and persistent employment responses. These papers establish the existence of monetary policy transmission to labor markets but do not address geographic heterogeneity in that transmission.

The geographic dimension of monetary policy has received less attention. Carlino and DeFina (1998) provide an early analysis showing that U.S. states with larger manufacturing sectors respond more strongly to monetary policy contractions, consistent with an interest-rate channel operating through credit-intensive industries. Their finding motivates the construction and manufacturing exposure measure used here. Relatedly, the shift-share design used in this paper draws on a large methodological literature originating with Bartik (1991), who shows that national industry employment growth interacted with local industry composition can be used to construct quasi-random local labor demand shocks. Goldsmith-Pinkham, Sorkin, and Swift (2020) formalize the identification assumptions behind shift-share instruments, noting that identification rests on the predetermined nature of the industry shares, precisely the assumption exploited here using the 2002 CBP composition.

This paper contributes to both literatures by combining the shift-share framework with a difference-in-differences design to study monetary policy heterogeneity at the county level. Unlike prior work that focuses on states or uses aggregate instruments, the county-level analysis provides sharper tests of the geographic channel and enables richer fixed-effects controls. The use of predetermined exposure shares from the CBP, rather than contemporaneous industry composition, addresses a key endogeneity concern in cross-sectional regressions of employment on industry shares.

3. Data and Methods

3.1 Data Sources

The analysis draws on three primary data sources. County-quarter employment counts come from the Quarterly Census of Employment and Wages (QCEW), published by the Bureau of Labor Statistics. The QCEW provides total private employment for all U.S. counties at a quarterly frequency, with near-universal employer coverage due to mandatory unemployment insurance reporting. The sample spans 2002 Q1 through 2019 Q4, excluding the 2008q4–2013q1 zero-rate window (see Section 4.3), yielding a panel of 155,757 county-quarter observations across 2,887 counties.

The exposure variable, each county's share of employment in construction and manufacturing, is constructed from the 2002 County Business Patterns (CBP) data published by the U.S. Census Bureau. The CBP provides industry-level employment counts at the county level, disaggregated by NAICS code. Construction (NAICS 23) and manufacturing (NAICS 31–33) are combined to form a single rate-sensitive employment share. Because this share is measured in 2002, before the sample begins, it is predetermined with respect to all subsequent federal funds rate decisions and county employment outcomes.

Federal funds rate data come from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. The quarterly change in the effective federal funds rate (ΔFFR_t) is the key monetary policy measure. Using changes rather than levels avoids issues of secular trend contamination and is consistent with standard practice in the monetary policy transmission literature.

Approximately 846 counties appear in the QCEW but not in the 2002 CBP and are excluded from the analysis. These counties are geographically dispersed with no distinctive industrial profile. For their omission to systematically bias results, excluded counties would need to differ in both industry composition and FFR employment response simultaneously — a joint condition that the spatial dispersion of missing observations makes implausible.

3.2 Descriptive Statistics

Table 1 reports summary statistics for the key variables in the analysis panel.

Variable	Mean	Std. Dev.	Min	Max
Log Employment	9.160	1.573	0.000	15.340
Change in Federal Funds Rate	0.005	0.395	-1.433	0.520
Exposure to IR Sensitive Industries	0.448	0.252	0.000	1.000
Observations	155,757 county-quarter observations			
Counties	2,887 counties			
Sample period: 2002 Q1–2019 Q4. Unit of observation: county-quarter.				
Exposure is the 2002 CBP construction and manufacturing employment share.				

3.3 Identification Strategy

The core identification challenge is that changes in the federal funds rate are endogenous to aggregate macroeconomic conditions. A simple regression of county employment on the FFR would conflate the policy effect with the underlying business cycle. The shift-share DiD design

addresses this by focusing on differential employment responses across counties with varying industry compositions, exploiting the fact that some counties are mechanically more exposed to interest rate changes due to their industrial structure.

The key insight is that while the level of the FFR is endogenous to aggregate conditions, its interaction with a predetermined county characteristic, the 2002 construction and manufacturing share, generates cross-sectional variation in exposure that is uncorrelated with time-varying county-level shocks, conditional on county and quarter fixed effects. The 2002 exposure measure predates the sample and cannot have been influenced by subsequent monetary policy decisions or employment outcomes.

The main regression specification is:

$$\ln(Emp_{ct}) = \beta_1 \Delta FFR_t + \beta_2 Exposure_c + \beta_3 (\Delta FFR_t \times Exposure_c) + \gamma_c + \delta_t + \varepsilon_{ct}$$

where $\ln(Emp_{ct})$ is the natural log of total private employment in county c in quarter t ; γ_c are county fixed effects that absorb all time-invariant differences across counties (geography, climate, historical industrial base); δ_t are quarter fixed effects that absorb all aggregate macroeconomic conditions common to all counties in a given quarter — including the level of the FFR itself, which is collinear with the quarter fixed effects by construction; $Exposure_c$ is the 2002 county construction and manufacturing employment share; and ΔFFR_t is the quarterly change in the effective federal funds rate. Standard errors are clustered at the county level to allow for arbitrary serial correlation within counties over time.

The coefficient β identifies the causal effect of monetary policy on employment for counties at the mean exposure level relative to counties at zero exposure, comparing changes in employment across rate-sensitive and rate-insensitive counties in quarters when the FFR changes versus quarters when it does not. Because quarter fixed effects absorb ΔFFR_t themselves, β is identified purely from the cross-sectional heterogeneity in employment responses across counties with different industry compositions — not from the aggregate time-series variation in rates.

3.4 Identifying Assumptions and Threats to Validity

The primary identifying assumption is that, absent rate changes, employment in high-exposure counties would have evolved in parallel with employment in low-exposure counties — the parallel trends assumption. The parallel trends assumption is fundamentally untestable as a counterfactual claim — we cannot observe what high-exposure counties would have done in the absence of rate changes. Instead, we provide supporting evidence through a pre-period visual inspection and a formal event study, both of which are presented below.

The main threat to validity is that the Federal Reserve's rate decisions are endogenous to current and expected macroeconomic conditions, including employment itself. This aggregate

endogeneity means the design does not estimate the causal effect of a *policy shock* per se, but rather the differential employment response to monetary policy cycles across counties with different industry compositions. The interaction design is robust to the aggregate endogeneity of the FFR because any aggregate employment effect of monetary policy is absorbed by the quarter fixed effects, leaving only the cross-sectional *differential* identified.

A secondary concern is that the 2002 exposure measure might be correlated with county characteristics that also predict differential employment trends. For example, manufacturing-heavy counties may have been on a secular decline independent of monetary policy. The county fixed effects absorb level differences, but correlated differential trends remain a concern. The event study pre-trend tests below address this directly.

4. Results

4.1 Main Specification

Table 2 reports the main regression results. The contemporaneous specification yields an interaction coefficient of 0.020 (SE = 0.004, $p < 0.01$), indicating that a 1 percentage point increase in the FFR is associated with approximately 2.0 percent higher employment in a county with all employment in construction and manufacturing, relative to a county with none. Evaluated at the sample mean exposure of 0.448, the implied differential is approximately 0.9 percent. This effect is precisely estimated and highly statistically significant.

To assess the economic magnitude, consider a county at the 75th percentile of exposure (approximately 0.63) versus a county at the 25th percentile (approximately 0.26). A one standard deviation FFR change (0.40 percentage points) would produce a differential employment change of approximately $0.020 \times 0.40 \times (0.63 - 0.26) \approx 0.30$ percent of county employment. While modest in any single quarter, these effects compound over multi-quarter rate cycles.

Table 2: Monetary Policy and County Employment

	(1) Baseline	(2) Distributed Lag
Rate Change $\times$ Exposure (t)	0.020*** (0.004)	-0.005 (0.007)
Rate Change $\times$ Exposure (t-1)		0.007 (0.005)
Rate Change $\times$ Exposure (t-2)		0.012 (0.009)
Rate Change $\times$ Exposure (t-3)		0.025*** (0.006)
Rate Change $\times$ Exposure (t-4)		0.004 (0.014)
Cumulative (t through t-4)		0.042*** (0.011)
County FE	Yes	Yes
Quarter FE	Yes	Yes
Clusters	2,887	2,887
Observations	155,757	132,674
R^2	0.993	0.993
Within R^2	0.000	0.001

Robust standard errors clustered at the county level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Cumulative effect estimated via `lincom`. Distributed lag

specification loses four quarters of observations due to lag construction.

4.2 Distributed Lag Specification

Column (2) of Table 2 presents the distributed lag specification, which allows employment to respond with delays to monetary policy changes. The contemporaneous interaction ($t = 0$) is close to zero and statistically insignificant ($\beta = -0.005$, $SE = 0.007$), suggesting little immediate differential response. The effect builds gradually: the lag-1 and lag-2 coefficients are positive but not statistically significant at conventional levels. The dominant effect materializes at lag 3—approximately 9 months after the rate change—with a coefficient of 0.025 ($p < 0.01$). The lag-4 coefficient returns toward zero ($\beta = 0.004$, $SE = 0.014$).

The cumulative four-quarter effect, the sum of all five interaction coefficients, is 0.042 ($SE = 0.011$, $p < 0.01$), indicating that over a full year, a one percentage point FFR increase is associated with approximately 4.2 percent greater employment growth in a county with all employment in construction and manufacturing, relative to a county with none. Evaluated at the sample mean exposure of 0.448, the implied cumulative differential is approximately 1.9 percent. This distributed lag pattern is economically sensible: construction and manufacturing projects

involve planning lags, contracts, and capital commitments that take several quarters to materialize into employment changes.

The distributed lag specification is the headline specification for this paper, as it more fully captures the dynamic employment response to monetary policy and avoids attributing contemporaneous quarter employment effects that manifest with a delay. The nine-month lag at the peak effect is consistent with the planning and production cycles of the construction and manufacturing sectors.

4.3 Parallel Trends

Figure 1 presents a visual parallel trends test, plotting average log employment over time for counties above and below the median of the exposure distribution. The two series track closely throughout the sample, with no systematic pre-existing divergence, providing visual support for the assumption of parallel trends. The level separation between the series reflects that high-exposure counties tend to have different population sizes than low-exposure counties. This difference is absorbed by county fixed effects and does not threaten identification.

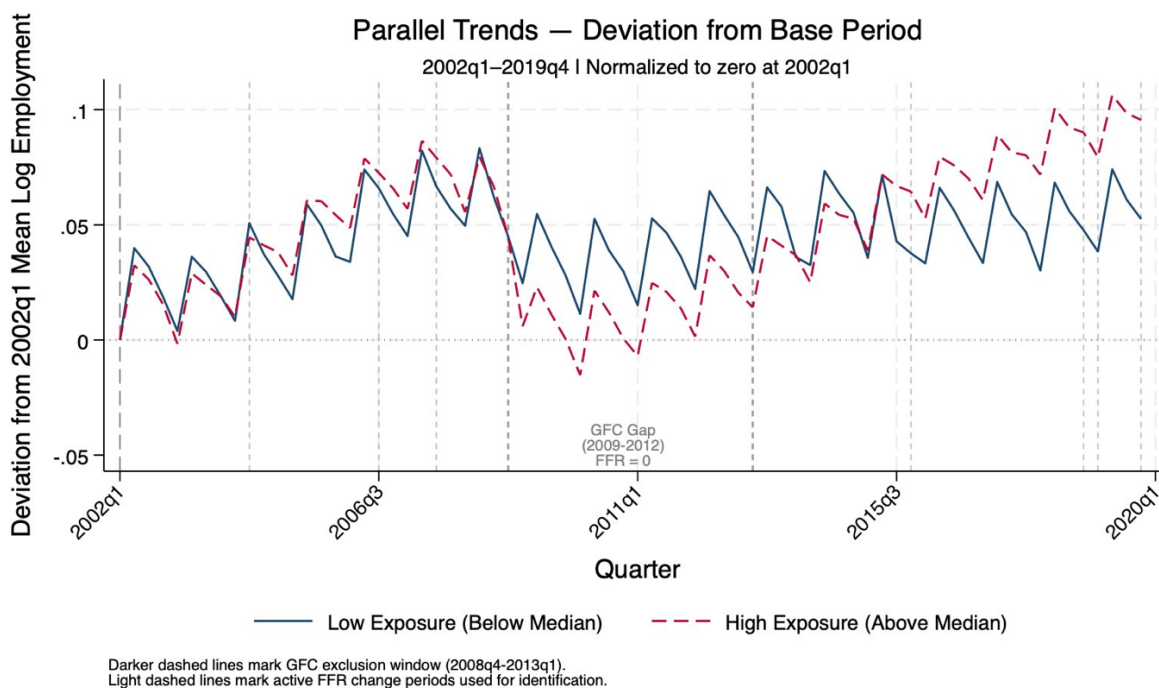


Figure 1: Parallel Trends — High vs. Low Exposure Counties, 2002–2019

Figure 2 extends this analysis to all four exposure quartiles, confirming that the four groups evolve in parallel throughout the sample period despite differing in their levels of employment. The absence of differential pre-trends across quartiles provides additional support for the identifying assumption.

Both figures include the 2009–2012 period for completeness, marked by dashed vertical lines. The federal funds rate was effectively zero throughout this window — a consequence of the Federal Reserve’s emergency response to the Global Financial Crisis rather than a policy decision of the kind studied here. Because there is no meaningful variation in the FFR during this period, these quarters contribute no identifying variation to the regression and are excluded from the analysis. The GFC window is also sufficiently distant from both the preceding hiking cycle (2004–2006) and the subsequent normalization cycle (2015–2018) to avoid biasing estimates in either sub-period.

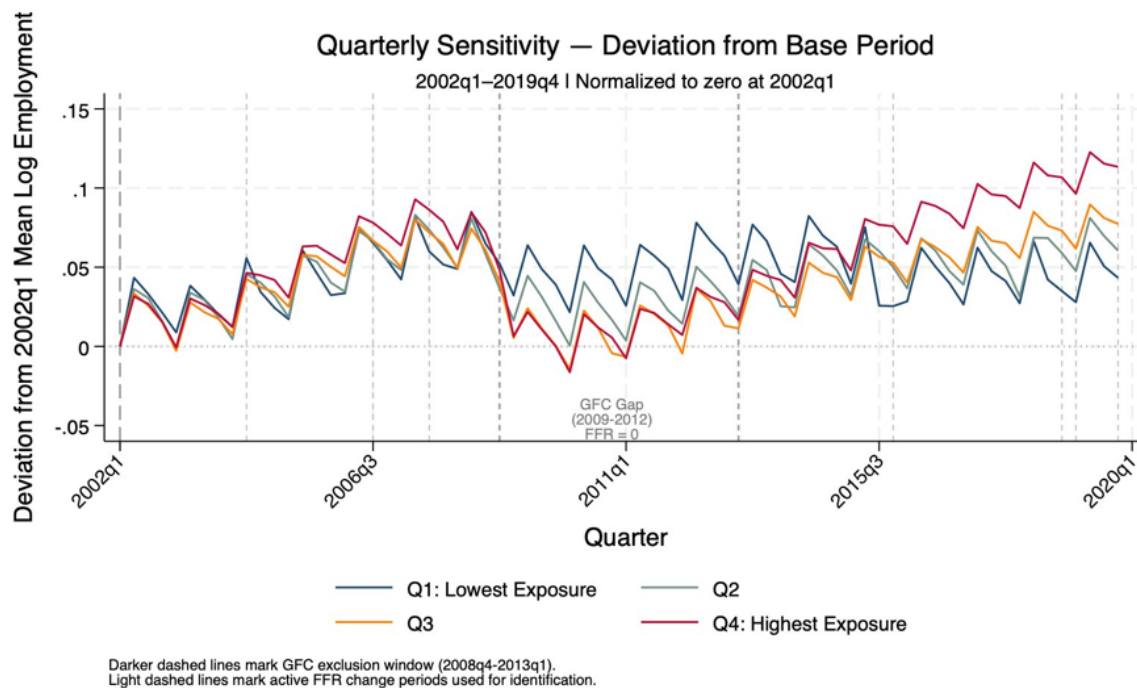


Figure 2: Quarterly Employment Trends by Exposure Quartile, 2002–2019

4.4 Event Study

Figure 3 presents an event study estimating the employment response to FFR changes in the lead and lag quarters, using a binary high-exposure indicator (above-median exposure). The $t = 0$ estimate is positive and statistically significant. The remaining post-period coefficients are small and statistically indistinguishable from zero, consistent with the effect being concentrated at the contemporaneous quarter in the binary exposure specification.

Only one pre-period lead (−4) is statistically significant; leads −3 and −2 are not distinguishable from zero, and lead −1 is the omitted reference period. A single significant pre-period coefficient, with no directional drift across leads, is more consistent with noise than with a structural divergence between high- and low-exposure counties. A genuine violation of parallel trends would manifest as consistent drift in one direction before $t = 0$; what the figure shows

instead is a single isolated lead with no discernible trajectory. The more natural interpretation is Fed endogeneity: the Federal Reserve reacts to short-run labor market fluctuations in rate-sensitive industries, generating a mechanical correlation between future rate changes and current employment in high-exposure counties. This is a recognized challenge in shift-share applications to monetary policy, and it does not undermine the main findings, which are identified from contemporaneous and near-term variation where this concern is far less acute.

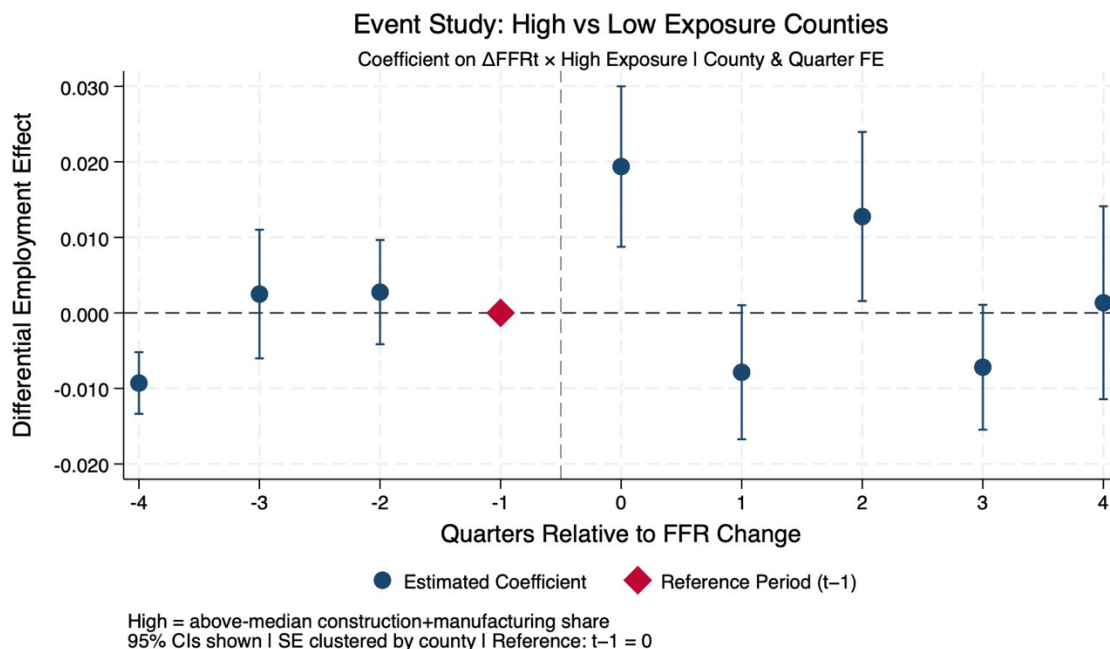


Figure 3: Event Study — Lead and Lag Effects of ΔFFR_t on Log Employment

5. Robustness Checks

5.1 Alternative Base-Year Exposure Measures

The primary exposure variable uses the 2002 CBP industry composition, the earliest available year in the sample. One concern is that the 2002 composition may reflect specific features of the pre-recession economy that do not generalize to the full sample period. Table 3 addresses this by repeating the main specification using exposure measures constructed from the 2003, 2013, and 2014 CBP — the latter two covering the post-recession period when the exposure composition may better reflect the industrial structure relevant to the 2013–2019 observations.

Across all four exposure specifications, the interaction coefficient is positive, statistically significant at $p < 0.01$, and quantitatively stable: the CBP 2002 baseline estimate of 0.020 changes only slightly to 0.023 (CBP 2003), 0.019 (CBP 2013), and 0.019 (CBP 2014). The tight clustering of estimates around 0.019–0.023 across base years spanning 12 calendar years, including both pre-

and post-recession periods, strongly suggests that the result is not an artifact of the year chosen for exposure construction, nor of changes in industrial structure over the sample period.

Table 3: Robustness Checks

	(1)	(2)	(3)	(4)	(5)
	CBP 2002	CBP 2003	CBP 2013	CBP 2014	$\Delta \ln(emp)$
Rate Change $\times$ Exposure	0.020*** (0.004)	0.023*** (0.005)	0.019*** (0.004)	0.019*** (0.004)	0.007*** (0.001)
County FE	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes
Clusters	2,887	2,834	2,810	2,811	2,887
Observations	155,757	152,895	151,684	151,738	155,757

Robust standard errors clustered at the county level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Cols (1)–(4): outcome is log total employment. Col (5): outcome is quarter-on-quarter log employment change. All exposure measures use construction + manufacturing share from the indicated CBP year.

5.2 Employment Growth Rate Outcome

Row (5) of Table 3 re-estimates the main specification, replacing log employment levels with quarterly employment growth ($\Delta \ln emp$) as the outcome variable. Using growth rates rather than levels provides a more direct measure of the flow of employment response to monetary policy changes. It sidesteps many concerns about non-stationarity in county employment levels over the 18-year sample.

The interaction coefficient in the growth rate specification is 0.007 (SE = 0.001, $p < 0.01$). The smaller magnitude compared to the levels specification is expected: the levels regression accumulates the employment response over time, while the growth rate specification captures the single-quarter differential. Scaled by the mean quarterly FFR change and a typical difference in exposure across quartiles, the growth rate estimate implies a differential quarterly employment growth of approximately 0.05 percentage points per unit FFR change — economically small but precisely estimated and consistent in sign with the main result.

5.3 Placebo Test

To probe the parallel trends assumption more directly, I construct a placebo test by replacing the contemporaneous change in the federal funds rate with its future leads. Specifically, I interact with one-through four quarters-ahead rate changes using county exposure and estimate whether these forward-looking interactions predict current employment outcomes. If counties are

not anticipating future monetary policy, a reasonable assumption given that firms and workers have no reliable foreknowledge of Fed decisions, all lead coefficients should be near zero.

Table 4: Placebo Test: Leads of Monetary Policy Shock

	(1) Lead Specification
Contemporaneous interaction	0.053*** (0.0095)
Lead 1 quarter	0.013 (0.0108)
Lead 2 quarters	-0.007 (0.0072)
Lead 3 quarters	0.015*** (0.0057)
Lead 4 quarters	-0.026*** (0.0096)
County FE	Yes
Quarter FE	Yes
Clusters	2,887
Observations	132,674

Robust standard errors clustered at the county level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 4 reports the results. The first two leads are small and statistically insignificant, which is reassuring. Leads 3 and 4, however, are significant and alternate in sign (+0.015, -0.026), a pattern that warrants some discussion. Anticipation is not a plausible explanation here: it strains credibility that local employers in construction and manufacturing would systematically adjust hiring three to four quarters before a rate change, which they cannot observe. A more natural reading is that the Federal Reserve itself responds to labor market conditions in these industries, creating a mechanical correlation between future rate changes and current employment in high-exposure counties. This is a well-recognized challenge in shift-share applications to monetary policy. Importantly, it does not undermine the main findings, which are identified in contemporaneous and near-term variation, where this reverse-causality concern is far less acute.

5.4 Interpretation of Robustness

Taken together, the robustness checks support three conclusions. First, the positive sign of the main interaction coefficient is not an artifact of the specific base year used to construct the exposure measure: it holds across exposure measures spanning 12 years, from pre-recession to post-recovery periods. Second, the result is not driven by a particular modeling choice for the outcome variable: both employment levels and employment growth rates yield consistent

inferences. Third, the magnitudes are stable enough across specifications to provide confidence in the quantitative interpretation of the main result.

6. Conclusion

This paper estimates the heterogeneous employment effects of federal funds rate changes across U.S. counties using a shift-share difference-in-differences design. Exploiting variation in predetermined construction and manufacturing employment shares as an exposure measure, the analysis finds that counties with greater pre-existing concentrations of rate-sensitive industries experience systematically higher employment growth during periods of monetary policy tightening. The main contemporaneous estimate is 0.020 ($p < 0.01$), and the cumulative four-quarter distributed lag effect is 0.042 ($p < 0.01$), with the dominant effect materializing at a nine-month lag. These results are robust to alternative base-year exposure measures and alternative outcome specifications.

The positive sign of the interaction coefficient is consistent with cyclical amplification: the Federal Reserve typically raises rates during economic expansions, and construction and manufacturing counties benefit disproportionately from boom conditions. This interpretation is consistent with the broader monetary policy transmission literature, which finds that rate-sensitive sectors amplify the cyclical component of aggregate monetary policy movements.

Several limitations warrant discussion. The design identifies differential exposure effects rather than the causal impact of a monetary policy shock: the aggregate FFR remains endogenous to macroeconomic conditions, and the interaction only isolates the cross-sectional heterogeneity in response. The event study reveals that lead -4 is statistically significant, suggesting that pre-existing employment dynamics in rate-sensitive counties may partially contaminate the identifying variation, consistent with the Fed reacting to employment conditions in these sectors. A single significant lead with no directional drift across the remaining pre-period coefficients makes a systematic pre-trend unlikely, but the pre-trend evidence is not entirely clean.

Three natural extensions could strengthen this analysis. First, adding county-specific linear time trends as an additional fixed effect would partially address correlated trends between industrial composition and employment trajectories. Second, using a Bartik-style national industry shift instrument — replacing the raw FFR change with a shift constructed from industry-level employment growth — would more cleanly isolate variation in local labor demand. Third, disaggregating the exposure measure into construction and manufacturing separately would test whether the two sectors respond differently to monetary policy, refining the mechanism.

Bibliography

- Bartik, Timothy. “Who Benefits from State and Local Economic Development Policies? Policies?,” 1991.
https://research.upjohn.org/cgi/viewcontent.cgi?article=1093&context=up_press.
- Bianchi, Francesco, Yuriy Gorodnichenko, David Romer, John Leahy, Serena Ng, and Tara Sinclair. “Are the Effects of Monetary Policy Shocks Big or Small?” April 2011.
https://economics.wm.edu/wp/cwm_wp90rev2.pdf.
- Blanchard, Olivier, Stanley Fischer, Christina Romer, and David Romer. “Chapter Title: Does Monetary Policy Matter? A New Test in the Spirit of Friedman and Schwartz Spirit O F Friedman and Schwartz.” *National Bureau of Economic Research* 4, no. 4 (1989): 0–262.
<https://www.nber.org/system/files/chapters/c10964/c10964.pdf>.
- Bureau, Labor Statistics. “Quarterly Census of Employment and Wages -- Data Files.” Bls.gov, 2018. <https://www.bls.gov/cew/downloadable-data-files.htm>.
- Bureau, US Census. “County Business Patterns Datasets.” Census.gov, December 2, 2024.
<https://www.census.gov/programs-surveys/cbp/data/datasets.html?text-list-cc4db501b6%3Atab=all#text-list-cc4db501b6>.
- Carlino, Gerald, and Robert DeFina. “The Differential Regional Effects of Monetary Policy.” *Review of Economics and Statistics* 80, no. 4 (November 1998): 572–87.
<https://doi.org/10.1162/003465398557843>.
- Federal Reserve Bank of St. Louis. “Effective Federal Funds Rate.” Stlouisfed.org, 2025.
<https://fred.stlouisfed.org/series/FEDFUNDS>.
- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. “Bartik Instruments: What, When, Why, and How,” 2018.
https://www.nber.org/system/files/working_papers/w24408/w24408.pdf.
- Nakamura, Emi, and Jón Steinsson. “High-Frequency Identification of Monetary Non-Neutrality: The Information Effect*.” *The Quarterly Journal of Economics* 133, no. 3 (January 29, 2018): 1283–1330.
- Romer, Christina D, and David H Romer. “A New Measure of Monetary Shocks: Derivation and Implications.” *American Economic Review* 94, no. 4 (August 2004): 1055–84.
<https://doi.org/10.1257/0002828042002651>.